# WETLAND DEGRADATION TECHNICAL REPORT

Multi-Temporal Spectral Transition Matrix, CVA Displacements, NDWI Dynamics & Fusion

| RUN ID | run_20260929_094943 | SOFTWARE VERSION | 1.0.0 |
|---|---|---|---|
| **ANALYSIS MODE** | wetland (two-date temporal) | **EXECUTION TIMESTAMP** | 2026-09-29T09:50:18 UTC |
| **OBSERVATION PERIOD** | 2023-08-14 → 2023-08-28 (14 days) | **ENVIRONMENT** | Python N/A \| N/A |

## 1. TECHNICAL OVERVIEW & GRID ALIGNMENT

| GRID SPECIFICATION | VALUE |
|---|---|
| **Spatial Reference (CRS)** | EPSG:32633 |
| **Reference Grid Dimensions** | 100 rows × 100 cols |
| **Pixel Resolution** | 10.00 m × 10.00 m |
| **Bounding Extent [W, S, E, N]** | [500000.00, 4999000.00, 501000.00, 5000000.00] |
| **Analysis Footprint Area** | 100.00 ha (10,000 valid pixels) |
| **Wetland Boundary Source** | Project-provided wetland boundary |

## 2. POST-CLASSIFICATION TRANSITION ANALYSIS

Every valid pixel is classified into one of 16 discrete state transitions between T1 and T2. Classes: 0=Background (B), 1=Severe Stress (S), 2=Moderate Stress (M), 3=Healthy (H):

| TRANSITION CATEGORY | PIXELS | AREA (HA) | SHARE (%) | ECOLOGICAL INTERPRETATION |
|---|---|---|---|---|
| **Stable Landscape** | 10,000 | 100.00 | 100.00% | Canopy condition remained unchanged between observation dates. |
| **Vegetation Degradation** | 0 | 0.00 | 0.00% | Transition from higher to lower health state (H→M, H→S, M→S). |
| **Vegetation Recovery** | 0 | 0.00 | 0.00% | Transition from lower to higher health state (S→M, S→H, M→H). |
| **Canopy Loss (Veg → Bg)** | 0 | 0.00 | 0.00% | Canopy disappearance into soil, water, or bare surface. |
| **Emergence (Bg → Veg)** | 0 | 0.00 | 0.00% | Vegetation emergence over formerly bare or water surface. |

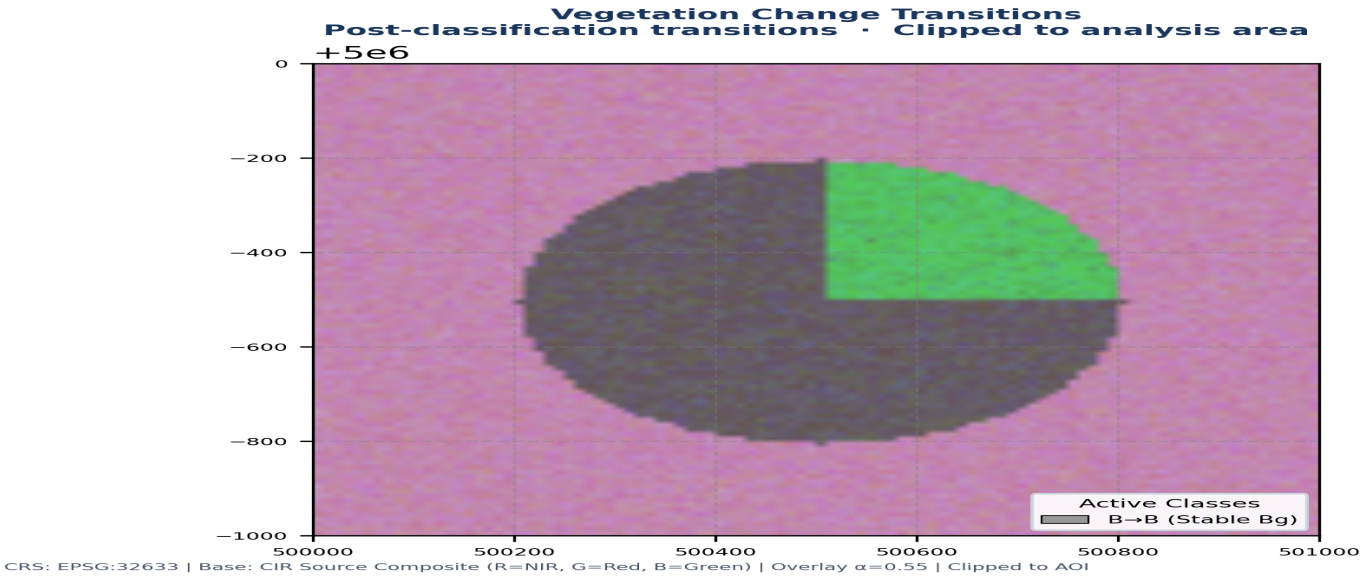



*Figure 1: Classified 16-State Transition Cartographic Plate*

## 3. CONTINUOUS CHANGE VECTOR ANALYSIS (CVA) & SPECTRAL SHIFTS

| METRIC | FORMULA / DESCRIPTION | VALUE |
|---|---|---|
| **Mean CVA Magnitude ‖ΔV‖** | `sqrt(ΔNDVI² + ΔGNDVI² + ΔNDRE²)` | 0.4641 |
| **Median CVA Magnitude** | 50th percentile of magnitude surface | 0.2902 |
| **95th Percentile Threshold** | P95 statistical outlier threshold | 1.2136 |
| **Mean ΔNDVI** | `NDVI(T2) − NDVI(T1) (canopy vigor shift)` | +0.3707 |
| **Mean ΔGNDVI** | `GNDVI(T2) − GNDVI(T1) (chlorophyll shift)` | +0.1186 |
| **Mean ΔNDRE** | `NDRE(T2) − NDRE(T1) (red-edge foliar shift)` | -0.0134 |

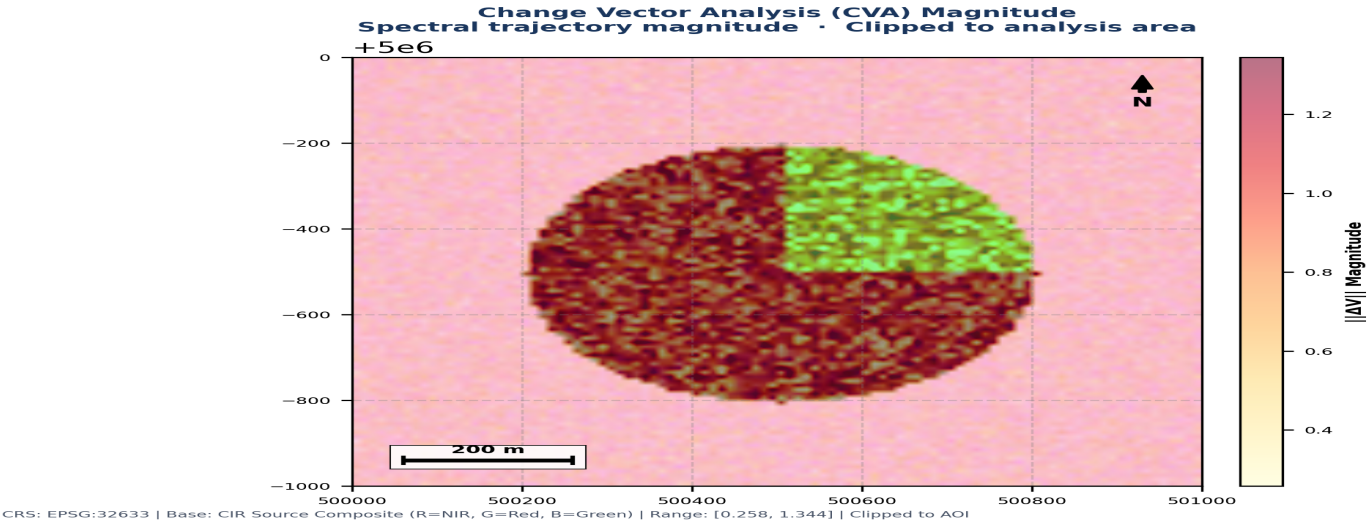



*Figure 2: Continuous Change Vector Analysis (CVA) Magnitude Plate*

## 4. HYDROLOGICAL EVIDENCE & SURFACE WATER DYNAMICS

| HYDROLOGICAL PARAMETER | VALUE |
|---|---|
| **NDWI Water Threshold** | `0.00 (Green − NIR) / (Green + NIR)` |
| **Surface Water Contraction (Loss)** | 0.00 ha |
| **Surface Water Expansion (Gain)** | 6.72 ha |
| **Net Water Surface Change** | +6.72 ha |

## 5. MULTI-EVIDENCE FUSION & DEGRADATION ASSESSMENT

| ASSESSMENT INDICATOR | SCIENTIFIC METRIC | STATUS / VALUE |
|---|---|---|
| **Hydro-Ecological Degradation** | 0.00 ha | Dual confirmation: vegetation degradation + surface water drying. |
| **Hydro-Ecological Recovery** | 0.00 ha | Dual confirmation: vegetation recovery + surface water expansion. |
| **Normalized Degradation Ratio (NDR)** | 0.0000 | Degraded canopy area normalized against total valid baseline vegetation. |
| **Severity-Weighted Degradation** | N/A | Weighted sum of transitions (H→S weight 1.0, H→M weight 0.5, M→S weight 0.5). |
| **Physical Wetland Depletion** | NOT_ASSESSED | Conservative safeguard: only confirmed when independent ground/boundary evidence corroborates. |

# 6. QUALITY ASSURANCE & PROVENANCE AUDIT

| AUDIT COMPONENT | STATUS | AUDIT DETAILS |
|---|---|---|
| **Mathematical Reconciliation** | PASS | Transition cell counts sum exactly to effective footprint (error < 0.001%). |
| **Spatial Alignment QA** | PASS | T1 and T2 raster footprints aligned to common grid without rotation errors. |
| **Radiometric Integrity** | PASS | CVA and spectral indices checked for numerical stability and inf/nan absence. |